Scalar Fourier Locality as the Rank-One Control for Periodic Hamiltonian Reduction

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Abstract

A finite-range matrix-valued periodic Hamiltonian need not yield a finite-range scalar field after spectral reduction to one ordered band. This report develops that mechanism as the rank-one control for the periodic-reduction program supporting the scientific lineage of Ragasa2026a. The exact parent is a four-state nearest-cell Hermitian field; the retained scalar Hamiltonian is its lowest ordered eigenvalue. Because scalar $U(1)$ conjugation is identically trivial, this family isolates interpolation, Fourier-tail truncation, direct Hermitian fitting, support/error selection, and numerical verification without the gauge-dependent represented-operator locality that appears at rank two. Exact finite-transform identities, Hermitian coefficient symmetry, error channel separation, Parseval closure, and the equality of uniform transform and least-squares routes are stated before numerical evaluation. Four construction meshes, a disjoint 1025-point evaluation inventory, sixteen symmetric supports, and two independent approximation routes provide multiple internal benchmarks. The sampled first-to-second gap is $0.37249453282422684E_{\mathrm{ref}}$. Complete-transform interpolation is near binary64 roundoff for construction sizes $N \geq 64$. Both approximation routes first satisfy the declared $10^{-8}E_{\mathrm{ref}}$ application threshold at range $R = 11$ on every construction mesh, and the passage remains sustained through $R = 15$. At production size $N = 256$, transform and direct-fit coefficient vectors agree within approximately 4.4×10^{-16}–$5.4 \times 10^{-16}E_{\mathrm{ref}}$ over the full range inventory. An independent numerical implementation reconstructs the typed result and authenticated presentation: all 725 named comparisons pass. These are provisional finite-control results, not continuum convergence, a global gap certificate, material validation, publication evidence, or a selected universal hopping range.

Keywords: Fourier locality; scalar band reduction; hopping truncation; periodic interpolation; Hermitian least squares; independent verification

1 Scientific question and evidence boundary

For the lowest ordered eigenvalue field of an exact finite-range periodic parent, how do finite interpolation, deliberate Fourier-support truncation, and direct Hermitian scalar fitting behave as construction resolution and retained hopping range are increased?

The question is deliberately narrower than a material tight-binding calculation. The parent is synthetic, one-dimensional, and exactly specified. The retained object is a scalar mathematical Hamiltonian on $\mathbb{C}$, not the four-state parent matrix on $\mathbb{C}^4$. The study compares scalar operators only after their state space, momentum convention, energy reference, and support convention have been aligned. It does not invoke Quantum ESPRESSO, Wannier90, a material crystal, or a continuum differential operator.

The present result was calculated once in memory, reconstructed by an independent software route, and rendered from a digest-authenticated projection. No retained execution directory, sealed package, release, archival deposit, or publication evidence record was created. Exact rational inputs and mathematical

identities are distinguished below from sampled binary64 observations. Passing software checks establish the implemented finite contract only.

This report is intended to be an upstream technical record for the scalar-control argument in the Ragasa2026a scientific program. A paper may distill its theory, benchmarks, and finite conclusions, but the paper is not the authority for the full calculation record. This statement does not modify the repository's current publication allocation ledger or any exact evidence manifest; those remain separate reviewed surfaces.

2 Motivation and literature position

Localized representations connect reciprocal-space electronic structure to real-space Hamiltonians. Maximally localized Wannier functions provide a systematic treatment for isolated and composite bands[5, 7, 4], while Wannier interpolation uses localized real-space information to reconstruct spectral quantities on dense reciprocal meshes[9]. The modern software ecosystem makes these ideas practical across a broad range of material calculations[3]. Questions of exact compact support and the choice of reduced tight-binding representations remain distinct from ordinary interpolation accuracy[6, 1].

This study does not reproduce a maximally localized Wannier function (MLWF) calculation and does not introduce a localization functional. Instead, it isolates a smaller mathematical fact needed before rank-two gauge comparisons can be interpreted: even when the ambient parent has nearest-cell matrix support, a scalar eigenvalue branch obtained by diagonalization can have a nontrivial Fourier tail. Consequently, "finite-range parent" and "finite-range reduced scalar band" are different statements.

The numerical structure is also a controlled instance of trigonometric interpolation and spectral truncation[8]. Complete discrete transforms, deliberately truncated series, and independently solved least-squares approximations answer different questions. Their discrepancies must be separated rather than collapsed into one error number. Because results near binary64 roundoff are sensitive to arithmetic and conditioning, numerical-accuracy claims are interpreted under standard floating-point cautions[2].

The rank-one family has a second role. In a scalar retained space, internal $U(1)$ conjugation cannot alter the represented Hamiltonian. It is therefore a null control for the gauge-sensitive matrix locality studied at rank two. Nontrivial Fourier decay in this report is caused by spectral reduction and finite approximation, not by an internal gauge attack.

3 Exact model and state spaces

3.1 Ambient parent and scalar retained operator

Let the ambient finite representation act on the standard basis of $\mathbb{C}^4$:

$$K(q) = T_0 + e^{2\pi i q} T_{+1} + e^{-2\pi i q} T_{+1}^{\dagger}, \qquad -\tfrac{1}{2} \le q < \tfrac{1}{2}. \tag{1}$$

The exact coefficient matrices, in units of the positive reference energy $E_{\mathrm{ref}} = 1$, are

$$T_0 = \begin{pmatrix} -6/5 & 3/50 & 0 & 0 \\ 3/50 & -7/20 & 0 & 0 \\ 0 & 0 & 1 & 1/25 \\ 0 & 0 & 1/25 & 7/4 \end{pmatrix}, \qquad T_{+1} = \begin{pmatrix} 3/20 & 0 & 3/25 & 0 \\ 0 & -1/10 & 0 & 1/10 \\ 0 & 0 & 1/20 & 0 \\ 0 & 0 & 0 & -1/25 \end{pmatrix}. \tag{2}$$

Every entry is exact rational data. Equation (1) is Hermitian for every real q, and its ambient Fourier support is exactly contained in $\{-1, 0, +1\}$.

Write the ordered eigenvalues as

$$\lambda_0(q) \le \lambda_1(q) \le \lambda_2(q) \le \lambda_3(q). \tag{3}$$

The retained scalar Hamiltonian is the mathematical field

$$H(q) = \lambda_0(q) \in \mathbb{R}, \tag{4}$$

acting on a one-dimensional retained state space. It is not a 1×1 block copied from $K(q)$, and it is not assumed to inherit the ambient nearest-cell support. The eigenvalue definition is backend-independent; a numerical eigensolver only evaluates it on declared finite point sets.

The first-to-second sampled gap is

$$\Delta(q) = \lambda_1(q) - \lambda_0(q). \tag{5}$$

A positive minimum over the declared samples qualifies those samples. It does not prove that $\Delta(q) > 0$ for every continuum momentum.

3.2 Why spectral reduction can create a tail

The characteristic polynomial $\det(K(q) - \lambda I) = 0$ has coefficients that are finite trigonometric polynomials in q, but an ordered root of that polynomial is generally not itself a finite trigonometric polynomial. Thus exact finite matrix support before diagonalization does not imply exact finite scalar support after selecting one eigenvalue branch. The present parent was chosen because it preserves compatibility with the rank-two controlled family while producing a nontrivial scalar tail. A direct one-orbital nearest-neighbor scalar parent was rejected as the principal benchmark because every range $R \geq 1$ would be exact and the support study would be nearly trivial.

3.3 Exact rank-one gauge boundary

Proposition 1 (Scalar conjugation invariance)**.** *For any scalar Hermitian field $H(q) \in \mathbb{R}$ and any phase $U(q) \in U(1)$,*

$$U(q)^\dagger H(q) U(q) = H(q). \tag{6}$$

Proof. Scalar multiplication commutes and $U(q)^\dagger U(q) = 1$, so $U(q)^\dagger H(q) U(q) = H(q) U(q)^\dagger U(q) = H(q)$. □

Equation (6) excludes separate pointwise-phase, global-phase, and known-inverse-phase locality channels. Emitting them as if they were distinct physical calculations would duplicate the same scalar field. This exact null result is the structural bridge to the rank-two study: a momentum-dependent internal unitary can redistribute matrix-valued hopping blocks only when the retained rank exceeds one.

4 Fourier representation and exact finite identities

4.1 Momentum and transform convention

For an even construction size N, use the centered half-open mesh

$$q_j = -\frac{1}{2} + \frac{j}{N}, \qquad j = 0, \ldots, N-1, \tag{7}$$

and the centered cyclic representatives

$$\mathcal{C}_N = \{-N/2, -N/2+1, \ldots, N/2-1\}. \tag{8}$$

For samples $H_j = H(q_j)$, define

$$t_r^{(N)} = \frac{1}{N} \sum_{j=0}^{N-1} H_j e^{-2\pi i q_j r}, \qquad r \in \mathcal{C}_N, \tag{9}$$

and the complete cyclic interpolant

$$I_N(q) = \sum_{r \in \mathcal{C}_N} t_r^{(N)} e^{2\pi i q r}. \tag{10}$$

The forward transform carries $1/N$ and the inverse carries no additional factor.

Proposition 2 (Construction-point closure)**.** *The complete cyclic interpolant satisfies*

$$I_N(q_j) = H_j \tag{11}$$

for every construction point, in exact arithmetic.

Proof. Substituting Equation (9) into Equation (10) yields the discrete character sum

$$N^{-1} \sum_{r \in \mathcal{C}_N} e^{2\pi i (q_j - q_\ell) r},$$

which equals one when $j = \ell$ modulo N and zero otherwise. □

4.2 Hermitian scalar coefficients

Because every H_j is real,

$$t_{-r}^{(N)} = \overline{t_r^{(N)}} \tag{12}$$

for representatives paired away from the cyclic Nyquist class. Ordinary symmetric supports are therefore

$$\mathcal{S}_R = \{-R, \ldots, R\}, \tag{13}$$

with $2R + 1 < N$ in this protocol. This strict inequality excludes the ambiguous negative Nyquist representative from every deliberately truncated model.

The transform-mediated truncation is

$$T_{N,R}(q) = \sum_{r=-R}^{R} t_r^{(N)} e^{2\pi i q r}. \tag{14}$$

It is a finite-range scalar hopping model. Its range is an ordinary lattice range, not a cyclic distance and not a material length.

4.3 Separated error channels

Let $\mathcal{E}_M$ be the disjoint evaluation inventory. Define

$$e_{\text{int}}(N) = \max_{q \in \mathcal{E}_M} |H(q) - I_N(q)|, \tag{15}$$

$$e_{\text{trunc}}(N, R) = \max_{q \in \mathcal{E}_M} |I_N(q) - T_{N,R}(q)|, \tag{16}$$

$$e_{\text{med}}(N, R) = \max_{q \in \mathcal{E}_M} |H(q) - T_{N,R}(q)|. \tag{17}$$

The pointwise residual telescopes exactly,

$$H - T_{N,R} = (H - I_N) + (I_N - T_{N,R}), \tag{18}$$

but maxima do not add. Only the bound

$$e_{\text{med}}(N, R) \le e_{\text{int}}(N) + e_{\text{trunc}}(N, R) \tag{19}$$

is implied. Threshold decisions use the total mediated error, never the truncation channel alone.

4.4 Independent direct Hermitian fit

The second approximation route solves, with equal positive weights,

$$\min_{a_0,\{a_r,b_r\}} \sum_{j=0}^{N-1} \left| H_j - a_0 - \sum_{r=1}^{R} (2a_r \cos 2\pi r q_j - 2b_r \sin 2\pi r q_j) \right|^2 . \tag{20}$$

The fitted coefficients are $d_0 = a_0$, $d_{+r} = a_r + ib_r$, and $d_{-r} = \overline{d_{+r}}$. The direct evaluation error is

$$e_{\mathrm{dir}}(N, R) = \max_{q \in \mathcal{E}_M} |H(q) - D_{N,R}(q)|. \tag{21}$$

Proposition 3 (Uniform-route equality)**.** *On the complete uniform construction mesh with equal weights and $2R + 1 < N$, the exact least-squares projection in Equation (20) selects the same retained Fourier coefficients as Equation (9) followed by symmetric truncation.*

Proof. The constant, cosine, and sine columns are mutually orthogonal discrete characters on the uniform mesh when their frequencies exclude the cyclic Nyquist class. Orthogonal projection onto their span therefore returns the corresponding discrete Fourier coefficients. The real Hermitian parameterization is precisely the conjugate-paired form of those coefficients. □

The implemented direct route nevertheless uses a separate least-squares owner and control flow. Its agreement with transform-mediated truncation is therefore both an analytic expectation and a numerical cross-route benchmark. For $R \geq 1$, the exact design condition number under the chosen column scaling is $\sqrt{2}$; the onsite-only $R = 0$ design has condition number one.

4.5 Parseval and finite perturbation controls

On the construction mesh, Parseval orthogonality gives

$$\frac{1}{N} \sum_{j=0}^{N-1} |H_j - T_{N,R}(q_j)|^2 = \sum_{r \in \mathcal{C}_N \setminus \mathcal{S}_R} |t_r^{(N)}|^2. \tag{22}$$

The square root of the right-hand side is the omitted-coefficient L^2 norm retained by the result. Equation (22) is a finite-transform identity, not a theorem about continuum exponential localization. Scalar absolute error is also the rank-one operator norm error, so no separate matrix-to-band perturbation inequality is needed in this control.

5 Protocol and benchmark design

5.1 Frozen finite inventories

The construction sizes are

$$N \in \{32, 64, 128, 256\}, \tag{23}$$

with $N = 256$ designated production. The evaluation inventory is

$$\widetilde{q}_i = -\frac{1}{2} + \frac{i + 1/257}{1025}, \qquad i = 0, \ldots, 1024. \tag{24}$$

All coordinates are formed as exact rational numbers before conversion to binary64. Exact rational set comparison establishes that Equation (24) is disjoint from every construction mesh.

Every integer range $R = 0, \ldots, 15$ is evaluated. The constraint $2R + 1 < 32$ ensures that every support is an ordinary conjugate-closed support for every construction mesh. The sampled gap threshold is $0.25E_{\mathrm{ref}}$, the application threshold is $10^{-8}E_{\mathrm{ref}}$, and the final adjacent-mesh refinement tolerance is 10^{-10} on the declared ordered channel vector.

5.2 Five complementary benchmarks

The protocol is not a single threshold check. It contains five complementary benchmarks:

1. **Structural null benchmark.** Proposition 1 proves that scalar internal phase rotations cannot alter the represented Hamiltonian.

2. **Complete-transform closure benchmark.** The full N-coefficient cyclic transform must reconstruct construction samples and satisfy Hermitian pairing near binary64 roundoff.

3. **Mesh-refinement benchmark.** Interpolation and every support-indexed channel are compared across $32 \to 64 \to 128 \to 256$.

4. **Independent-route benchmark.** Transform-mediated truncation is compared with an independently solved Hermitian least-squares model in coefficient and sampled spaces.

5. **Decision benchmark.** The first jointly passing range, sustained passage, support/error nondominance, and near-tie status are reconstructed independently.

These benchmarks separate mathematical structure, transform correctness, discretization behavior, algorithmic route agreement, and decision policy.

5.3 Qualification and decision rules

The sampled gap gate passes only when the minimum across the production construction mesh and common evaluation inventory is strictly above $0.25E_{\mathrm{ref}}$. The application decision at range R passes only when both e_{med} and e_{dir} are at or below $10^{-8}E_{\mathrm{ref}}$. The first passage must remain passing at every larger attained range. The production range/error mask compares $(2R+1, e_{\mathrm{med}})$; other diagnostics are not silently substituted as objectives.

No rule in this protocol selects a universal adequate range. The first-passing result belongs to this parent, these meshes, this error metric, and this tolerance.

5.4 Calculation, verification, and rendering routes

The producer uses PhysKit complex-Hermitian eigensystems, KSD finite-periodic transform and interpolation owners, and the Project Koios Optimization least-squares owner. It returns one immutable typed result containing the parent samples, transform coefficients, candidates, errors, norms, conditioning diagnostics, decisions, and refinement ledger.

The independent verifier does not invoke the production analyzer. It reconstructs the parent matrices and eigensystems, all transforms and fits, Parseval channels, sampled gap, refinement inventory, masks, passage decision, and visualization projection using direct NumPy and SciPy formulas. The verifier owns the exact admitted result bytes and derives the subject Secure Hash Algorithm 256-bit (SHA-256) identity from those bytes. The renderer accepts only passing verification and the exact authenticated projection; it calculates no new scientific values.

6 Finite results

6.1 Sampled gap and retained-band qualification

The minimum sampled first-to-second gap is

$$\Delta_{\mathrm{min}} = 0.37249453282422684E_{\mathrm{ref}}. \tag{25}$$

It occurs on the production construction inventory at index 128, $q = 0$. The value is above the declared $0.25E_{\mathrm{ref}}$ gate. This establishes only sampled qualification. It is not a global isolation certificate and carries no uncertainty interval between samples.

6.2 Complete-transform closure and interpolation

Table 1: Complete-transform and interpolation benchmark. Energy-valued columns use units of E_{ref}.

N	reconstruction max.	Hermiticity max.	interpolation max.	interpolation RMS	first pass	sustained
32	4.441×10^{-16}	1.310×10^{-18}	2.234057×10^{-11}	1.077042×10^{-11}	11	yes
64	4.449×10^{-16}	9.136×10^{-18}	8.882×10^{-16}	2.903×10^{-16}	11	yes
128	4.458×10^{-16}	1.676×10^{-17}	1.110×10^{-15}	3.052×10^{-16}	11	yes
256	6.670×10^{-16}	2.815×10^{-17}	8.882×10^{-16}	3.121×10^{-16}	11	yes

The $N = 32$ complete interpolant differs from the exact parent evaluation by $2.2340573835322175 \times 10^{-11} E_{\mathrm{ref}}$. For $N \geq 64$, the interpolation maximum is near binary64 roundoff. Complete construction reconstruction remains below $6.67 \times 10^{-16} E_{\mathrm{ref}}$ for all four meshes. These two statements answer different questions: transform closure checks the finite discrete transform on its own nodes, whereas interpolation compares that transform with newly evaluated parent values on a disjoint mesh.

The maximum changes in the full ordered refinement channel vector are $2.2339685656902475 \times 10^{-11}$, $4.17817502814292 \times 10^{-15}$, and $6.2211547825082476 \times 10^{-15}$ for the three adjacent mesh pairs. The final value passes the 10^{-10} refinement gate. The observed nonmonotone last digits are roundoff-scale behavior, not evidence of asymptotic order.

6.3 Threshold-adjacent route comparison

Table 2: Joint first-passage values at $R = 11$ for every construction mesh, in units of E_{ref}.

N	mediated total maximum	direct total maximum	coefficient defect	design condition
32	$9.1053783402 \times 10^{-9}$	$9.1053777851 \times 10^{-9}$	7.975×10^{-16}	$\sqrt{2}$
64	$9.1054125351 \times 10^{-9}$	$9.1054129792 \times 10^{-9}$	7.995×10^{-16}	$\sqrt{2}$
128	$9.1054123130 \times 10^{-9}$	$9.1054127571 \times 10^{-9}$	2.912×10^{-16}	$\sqrt{2}$
256	$9.1054123130 \times 10^{-9}$	$9.1054128681 \times 10^{-9}$	4.783×10^{-16}	$\sqrt{2}$

At $R = 10$, the production mediated and direct maxima are respectively $3.164490658402741 \times 10^{-8} E_{\mathrm{ref}}$ and $3.164490702811662 \times 10^{-8} E_{\mathrm{ref}}$, so both fail. At $R = 11$, both pass as shown in Table 2. Every larger attained range passes. The threshold crossing is therefore bracketed by an explicit failing predecessor and a passing selected range on every construction mesh.

6.4 Complete production range study

Table 3: Production $N = 256$ range diagnostics in units of E_{ref}. Displayed values are rounded; the typed result retains binary64 values.

R	e_{trunc}	e_{med}	e_{dir}	omitted L^2	coefficient defect L^2
0	2.969490×10^{-1}	2.969490×10^{-1}	2.969490×10^{-1}	2.094025×10^{-1}	4.441×10^{-16}
1	1.529255×10^{-3}	1.529255×10^{-3}	1.529255×10^{-3}	7.868300×10^{-4}	4.455×10^{-16}
2	4.702662×10^{-4}	4.702662×10^{-4}	4.702662×10^{-4}	2.415628×10^{-4}	4.462×10^{-16}

R	e_{trunc}	e_{med}	e_{dir}	omitted L^2	coefficient defect L^2
3	1.454887×10^{-4}	1.454887×10^{-4}	1.454887×10^{-4}	7.488650×10^{-5}	4.618×10^{-16}
4	4.475369×10^{-5}	4.475369×10^{-5}	4.475369×10^{-5}	2.309687×10^{-5}	4.498×10^{-16}
5	1.366150×10^{-5}	1.366150×10^{-5}	1.366150×10^{-5}	7.070482×10^{-6}	4.752×10^{-16}
6	4.136042×10^{-6}	4.136042×10^{-6}	4.136042×10^{-6}	2.146950×10^{-6}	4.673×10^{-16}
7	1.241321×10^{-6}	1.241321×10^{-6}	1.241321×10^{-6}	6.463664×10^{-7}	4.778×10^{-16}
8	3.691080×10^{-7}	3.691080×10^{-7}	3.691080×10^{-7}	1.928393×10^{-7}	4.717×10^{-16}
9	1.086649×10^{-7}	1.086649×10^{-7}	1.086649×10^{-7}	5.697590×10^{-8}	4.599×10^{-16}
10	3.164491×10^{-8}	3.164491×10^{-8}	3.164491×10^{-8}	1.665750×10^{-8}	4.590×10^{-16}
11	9.105412×10^{-9}	9.105412×10^{-9}	9.105413×10^{-9}	4.813878×10^{-9}	4.783×10^{-16}
12	2.584796×10^{-9}	2.584796×10^{-9}	2.584796×10^{-9}	1.373266×10^{-9}	4.830×10^{-16}
13	7.224646×10^{-10}	7.224646×10^{-10}	7.224648×10^{-10}	3.860166×10^{-10}	4.757×10^{-16}
14	1.982817×10^{-10}	1.982817×10^{-10}	1.982825×10^{-10}	1.066565×10^{-10}	5.409×10^{-16}
15	5.322764×10^{-11}	5.322764×10^{-11}	5.322831×10^{-11}	2.886786×10^{-11}	5.372×10^{-16}

The range-zero approximation retains only the mean and has error $2.969489539542949 \times 10^{-1} E_{\mathrm{ref}}$. Adding the first neighbor reduces the error by nearly two orders of magnitude. Thereafter the tail decreases regularly over the attained inventory. The final range-15 mediated error is $5.3227644514208805 \times 10^{-11} E_{\mathrm{ref}}$; it is not zero because the reduced scalar band is not a finite-range trigonometric polynomial on the evidence available here.

Every attained production support is nondominated under the declared $(2R + 1, e_{\mathrm{med}})$ ordering because error decreases strictly as complexity increases. No attained production supports form a near-tie under the mixed 10^{-12} rule. This describes the finite inventory; it does not extrapolate the decay law beyond $R = 15$.

7 Diagnostic interpretation

7.1 Coefficient decay, bandwidth, and curvature

The complete production coefficients are conjugate symmetric near numerical precision. The dominant magnitudes are $|t_0| = 1.2116915623409472 E_{\mathrm{ref}}$ and $|t_{\pm 1}| \approx 0.1480688936028328 E_{\mathrm{ref}}$. Magnitudes then fall from $5.295039809188736 \times 10^{-4} E_{\mathrm{ref}}$ at $|r| = 2$ to approximately $7.26 \times 10^{-11} E_{\mathrm{ref}}$ at $|r| = 15$. This observed decay explains why the first neighbor captures most of the bandwidth while the $10^{-8} E_{\mathrm{ref}}$ criterion requires a substantially longer tail.

At $R = 15$, the production truncated scalar bandwidth is $0.5915561172409006 E_{\mathrm{ref}}$, and the zone-center second derivative with respect to reduced momentum is $-11.2914491774051 08 E_{\mathrm{ref}}$. These are representation diagnostics of the finite scalar model. They are not physical velocity, effective mass, or material curvature because no reciprocal-length or particle-mass convention is supplied.

7.2 Parseval and real-valued closure

The largest production Parseval absolute residual across all ranges is $1.3877787807814457 \times 10^{-17} E_{\mathrm{ref}}^2$. The largest imaginary residual in the range-15 production interpolant is $8.414211492160626 \times 10^{-17} E_{\mathrm{ref}}$. Together with the Hermiticity defects in Table 1, these checks show that the finite transform, conjugate pairing, and omitted-power accounting close at the expected floating-point scale.

Parseval closure does not establish that the chosen support is scientifically adequate. It establishes that the omitted coefficient norm and construction-space residual are consistent under the declared finite transform.

7.3 Route agreement and what it rules out

Across the production range inventory, transform and direct-fit coefficient defects are approximately 4.4×10^{-16}–$5.4 \times 10^{-16} E_{\mathrm{ref}}$. The maximum production construction-sample route defect remains below $1.56 \times 10^{-15} E_{\mathrm{ref}}$. All direct designs are full rank; their condition numbers are one at $R = 0$ and approximately $\sqrt{2}$ for $R \geq 1$.

This agreement rules out a gross implementation disagreement between the discrete transform truncation and the independent Hermitian regression parameterization. It does not constitute independent experimental evidence: both routes use the same parent samples, binary64 arithmetic, and development environment.

8 Authenticated figures

Figures 1 and 2 are retained from the verifier-authenticated projection. They summarize the numerical record but do not own values independently of the typed result and verification.

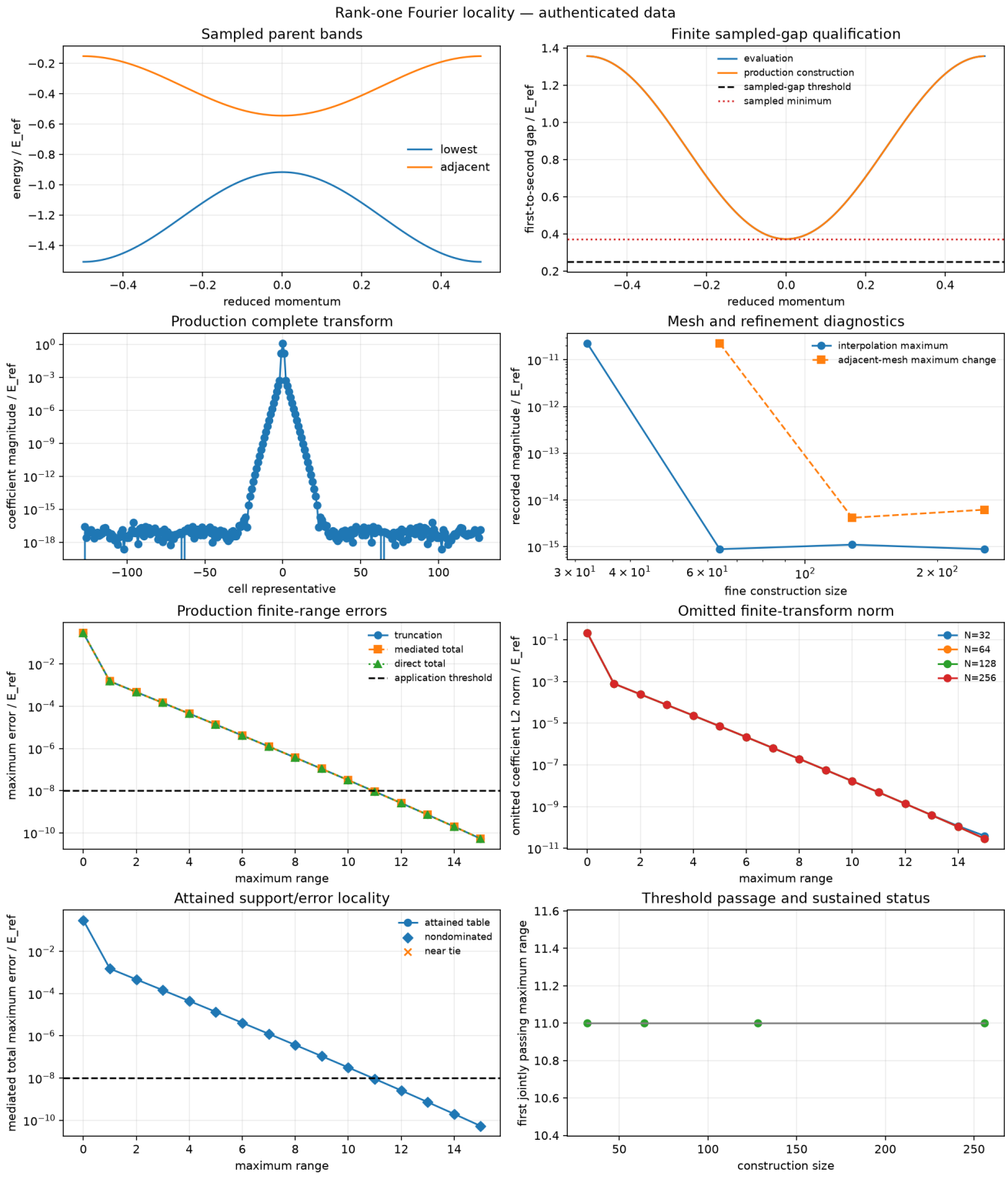


Figure 1: Authenticated rank-one Fourier-locality benchmark. The panels report the parent bands, sampled gap gates, complete coefficient decay, interpolation and adjacent-mesh refinement, production error channels, omitted norms, attained support/error decisions, and first sustained threshold passage. Lines between sampled points guide the eye and are not continuous-domain certificates.

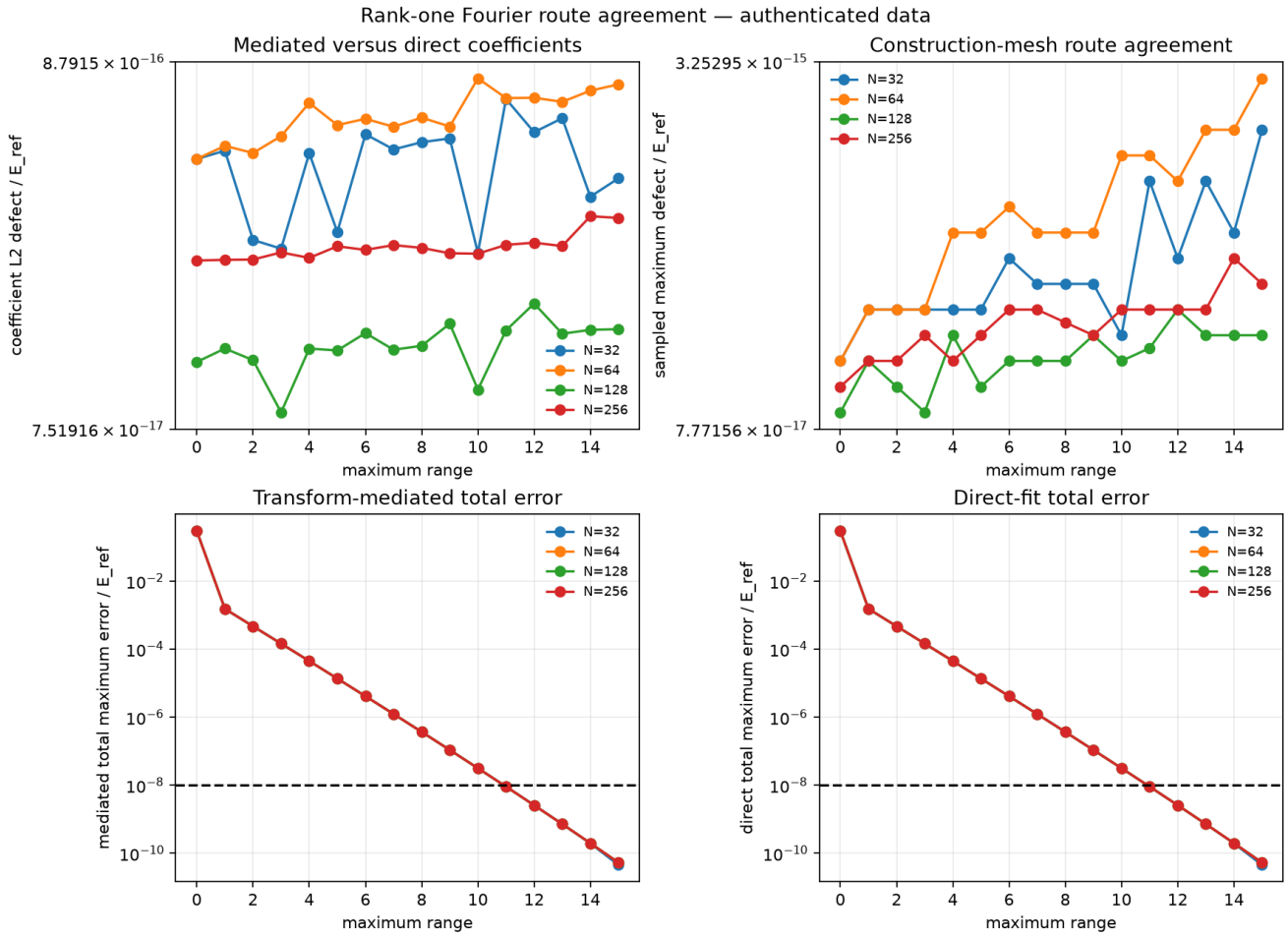


Figure 2: Authenticated independent-route benchmark. The upper panels show coefficient and construction-sample defects between transform-mediated truncation and the direct Hermitian fit. The lower panels compare total evaluation errors with the declared $10^{-8}E_{\mathrm{ref}}$ threshold.

9 Independent verification and authenticated presentation

The byte-owning verifier reconstructs the exact typed result from the same immutable bytes used for the subject digest. It independently regenerates:

- exact-rational mesh inventories and their disjointness;
- bound parent coefficients, Hermitian fibers, ordered eigensystems, and both sampled bands;
- complete transforms, coefficient pairing, construction reconstruction, and disjoint-mesh interpolation;
- every truncated and directly fitted candidate on construction and evaluation inventories;
- maximum and root-mean-square error channels, omitted norms, Parseval values, bandwidths, curvatures, imaginary residuals, design ranks, and condition numbers;
- coefficient and sampled route defects;
- sampled-gap minimizer, adjacent-mesh refinement ledger, nondominated and near-tie masks, first joint passage, and sustained-passage disposition; and
- every field in the authenticated visualization projection.

All 725 named observations passed. The maximum observed discrepancy was $9.672262990534364 \times 10^{-12}$; it remained within its applicable mixed tolerance. The maximum tolerance in the aggregate was $1.2691051231735601 \times 10^{-11}$. The result, verification aggregate, and visualization projection survive canonical byte-identical JavaScript Object Notation (JSON) round trips.

The independent route does not call the production analyzer or deserialize a historical campaign result. It shares binary64 arithmetic, NumPy/SciPy lineage, and the same local machine. It is therefore producer-independent software reconstruction, not an independent physical experiment, alternative precision study, or different-hardware reproduction.

10 Relation to the rank-two program and Ragasa2026a

The scalar control supplies three pieces of upstream scientific support.

First, it establishes the nontrivial baseline: spectral reduction of a nearest-cell four-state parent can produce a scalar Fourier tail even when internal gauge action is trivial. A rank-two claim that a momentum-dependent frame redistributes represented matrix locality must therefore separate ordinary spectral-reduction tails from genuine frame dependence.

Second, it qualifies the common numerical machinery—centered transforms, ordinary support conversion, disjoint evaluation, deliberate truncation, direct fitting, first-passage decisions, and independent reconstruction—without a non-Abelian alignment variable. Failures at rank two can then be diagnosed against a scalar null control rather than attributed immediately to gauge structure.

Third, it provides a complete error-versus-range record from which a paper can distill only the comparisons relevant to its argument. The technical report intentionally contains more derivation, channel definition, benchmark detail, and provenance than a journal manuscript should carry.

The present calculation is not automatically publication evidence for Ragasa2026a. Exact publication use requires the maintained allocation ledger and evidence manifests to bind the intended result bytes. The retained Mathieu1D scalar precursor and this controlled-four-state scalar family remain scientifically distinct; neither identity is silently substituted for the other.

11 Discussion

11.1 What the finite study demonstrates

Within the declared inventories, three conclusions are strongly supported. The reduced scalar band has a rapidly decreasing but nonzero resolved Fourier tail. Construction sizes $N \geq 64$ agree on the disjoint evaluation inventory near binary64 roundoff. Two mathematically equivalent but separately implemented approximation routes give the same first-passing support and agree in coefficients and samples near roundoff.

The first-passing value $R = 11$ should be read as a controlled decision, not an intrinsic localization length. If the application threshold, parent, norm, mesh, or support rule changes, the selected range may change. The table is more informative than the single selection because it preserves the complete finite tradeoff.

11.2 Why the rank-one result is scientifically useful

The exact $U(1)$ null result prevents a conceptual category error. Scalar Fourier locality can be nontrivial without being gauge-sensitive. At rank two, by contrast, pointwise unitary similarity preserves complete spectra while changing the matrix field whose Fourier coefficients are truncated. The scalar study therefore identifies which parts of the later methodology are common numerical infrastructure and which require a higher-rank state-space action.

11.3 Why the evidence remains provisional

The evaluation mesh is dense relative to the construction meshes, but it is still finite. No analytic strip, derivative bound, interval enclosure, or arbitrary-precision study converts sampled gap or error statements into continuum theorems. The refinement ladder shows finite stability under its declared channels; it does not quantify model or roundoff uncertainty. No external code or retained execution reproduces the result.

12 Provisional provenance

The development calculation used branch `work/periodic1d-rank1-control-v3` at base revision `bc1601f3a50dda19b994eea474ca522f2d836c26` in a dirty worktree. Runtime on the local development machine was approximately 1.12 seconds for calculation, 0.19 seconds for independent verification, and 0.88 seconds for rendering. These timings are descriptive, not benchmarks.

Exact SHA-256 identities from the non-retained development calculation were:

- configuration bytes: `5c9beb47eefe9749a249a74545a25c7ef8d3dd6eafec658d23dfbee574ace173`;
- result bytes: `3f24fbf6bd6d85040e98481b0487407932f43a70dd04690417396a50797fad2b`;
- verification bytes: `69232918f32f7006df0a5818925c35ff1462d4120448692aebdfc4a1489fd366`;
- visualization-data bytes: `5fd7fb9bf56c18cf112ae1060b672d1990c5cfc97661321eac4c735fdb816f3a`;
- locality report PNG: `df7032249383521b91012d18ef83025a72e6333b387e6b97b9acc898b2fdff62`;
- route-agreement report PNG: `c95f9ab1289baefffba637197d40218324f46c9b2b18476a8bccf2067352cf07`.

SHA-256 establishes exact byte identity only. It does not establish scientific validity, authorship, publication, archival retention, release, or sealing. No retained result or verification file for this family exists in the repository.

13 Limitations and nonclaims

This report does not establish:

- a continuum limit or a globally certified isolated band;
- an exponential-decay rate or a compact-support theorem for the scalar band;
- uncertainty quantification or a forward-error bound for every reported digit;
- material applicability, physical hopping lengths, effective masses, or velocities;
- higher-rank gauge behavior, topology, or a Wannier-spread minimum;
- adequacy of range 11 for another parent, tolerance, norm, or observable;
- publication acceptance, retained evidence, release, or sealing; or
- a Project Koios–Formal certificate.

These exclusions are part of the scientific result boundary, not boilerplate. Each would require additional definitions, controls, or protected actions.

14 Conclusions

The rank-one controlled-four-state study is a substantive scientific control rather than a trivial scalar example. Its exact contribution is to separate three mechanisms: finite-range ambient matrix structure, spectral reduction to a scalar eigenvalue field, and finite approximation of that scalar field. Internal scalar gauge action is exactly inert, but the reduced band develops a resolved Fourier tail.

The finite calculation then exercises a complete methodology: four construction meshes, a disjoint evaluation inventory, sixteen ordinary supports, transform and direct-fit routes, gap and refinement gates, full error-channel accounting, decision reconstruction, and authenticated presentation. The sampled gap passes, complete interpolation is roundoff-scale for $N \geq 64$, both routes first pass at $R = 11$, and 725 independent comparisons pass.

This complete record is the level from which publication prose should be distilled. A paper may emphasize the scalar null boundary and selected range comparisons, but it should not be the sole location of the supporting theory, benchmarks, diagnostics, and provenance.

Table 4: Declared controls for the provisional finite calculation.

Control	Value
Parent identity	`controlled-four-state-parent.v1`
Ambient / retained dimensions	4/1
Construction sizes	$32, 64, 128, 256$
Production construction size	256
Evaluation size	1025
Evaluation shift	1/257 of one evaluation step
Ranges	$0, 1, \ldots, 15$
Sampled gap threshold	$0.25E_{\mathrm{ref}}$
Application threshold	$10^{-8}E_{\mathrm{ref}}$
Energy absolute tolerance	$10^{-12}E_{\mathrm{ref}}$
Relative tolerance	10^{-12}
Squared-energy tolerance	$10^{-12}E_{\mathrm{ref}}^2$
Dimensionless tolerance	10^{-12}
Final refinement tolerance	10^{-10}

A Frozen control inventory

B Interpretation map

Object	What it establishes	What it does not establish
Exact parent coefficients	Finite matrix family and units	Material parametrization
Scalar $U(1)$ proposition	No internal scalar gauge redistribution	Higher-rank gauge behavior
Complete reconstruction	Discrete transform closure	Evaluation accuracy
Disjoint interpolation	Sampled off-grid agreement	Continuum convergence
Omitted L^2 norm / Parseval	Finite coefficient-power accounting	Application adequacy
First joint passage	Decision under declared threshold	Universal hopping range
Independent verification	Separate software reconstruction	Independent physical experiment
SHA-256 digest	Exact byte identity	Scientific validity or sealing

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